Spatial Rads Update

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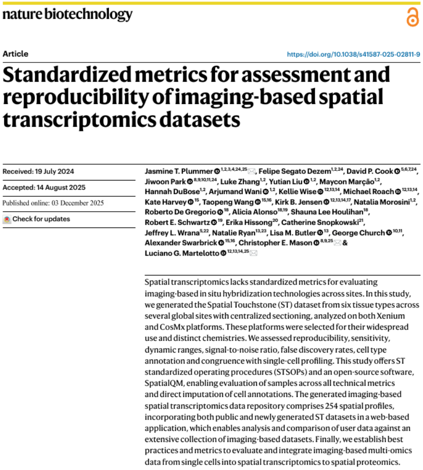
Method Updates and Changes

New Input Level QC

Plummer *et al.*, *Nature Biotechnology* (2025)

- Per-cell data quality across experiments
- Per-cell data quality across cohorts
- Bias
- Between replicate concordance
- Cell neighborhood purity


nature biotechnology

Article

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Standardized metrics for assessment and reproducibility of imaging-based spatial transcriptomics datasets

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Check for updates

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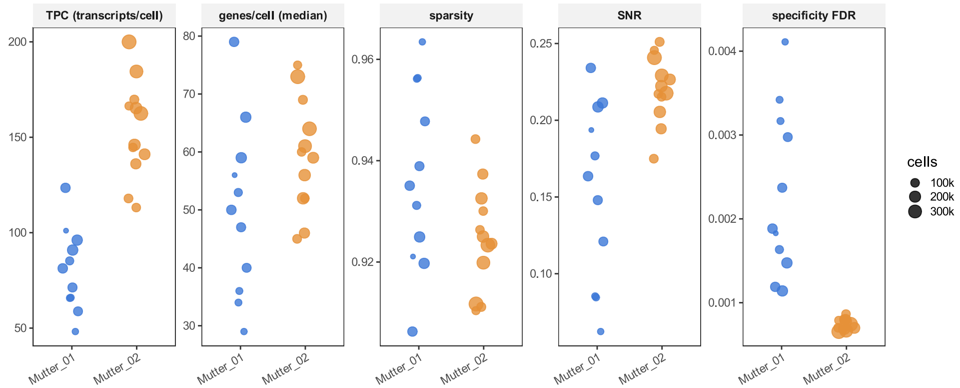
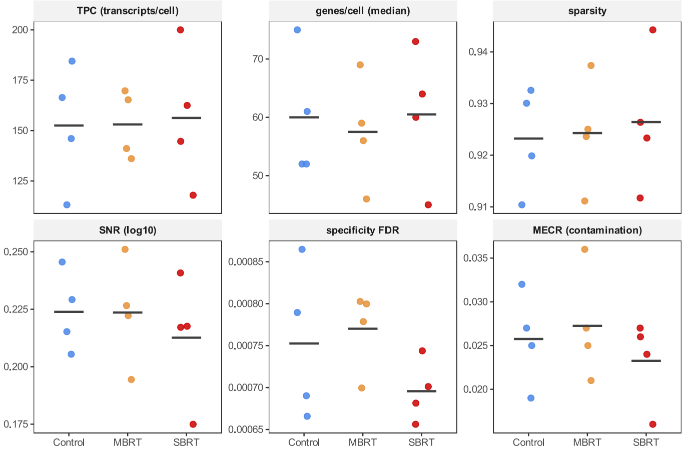
Spatial transcriptomics lacks standardized metrics for evaluating imaging-based in situ hybridization technologies across sites. In this study, we generated the Spatial Touchstone (ST) dataset from six tissue types across several global sites with centralized sectioning, analyzed on both Xenium and CosMx platforms. These platforms were selected for their widespread use and distinct chemistries. We assessed reproducibility, sensitivity, dynamic ranges, signal-to-noise ratio, false discovery rates, cell type annotation and congruence with single-cell profiling. This study offers ST standardized operating procedures (STSOPs) and an open-source software, SpatialQM, enabling evaluation of samples across all technical metrics and direct imputation of cell annotations. The generated imaging-based spatial transcriptomics data repository comprises 254 spatial profiles, incorporating both public and newly generated ST datasets in a web-based application, which enables analysis and comparison of user data against an extensive collection of imaging-based datasets. Finally, we establish best practices and metrics to evaluate and integrate imaging-based multi-omics data from single cells into spatial transcriptomics to spatial proteomics.

Per-cell Data Quality Across Experiments



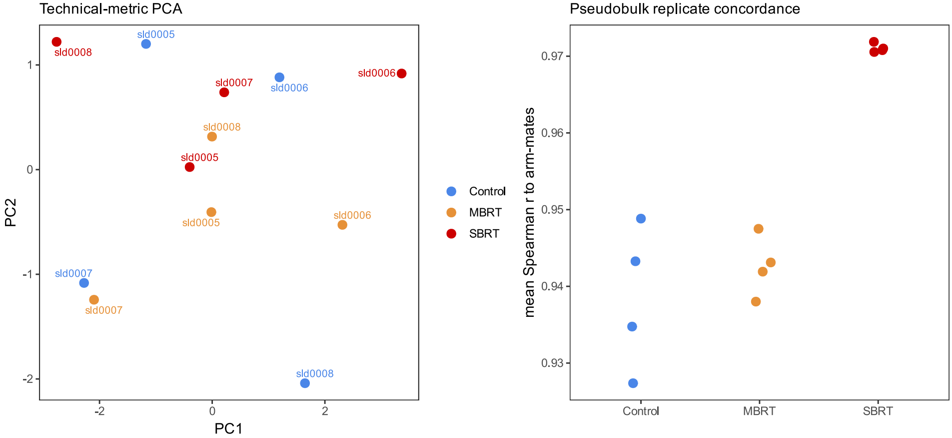
Per-sample metrics of CosMx capture across experiments. Sparsity, zeros of cell x gene counts; SNR, signal to noise; specificity FDR, expected false positives by background probe count. Overall M_02 has more transcripts per cell.

Per-cell Data Quality Across M_02 Cohorts



Per-sample metrics of CosMx capture across cohorts in Mutter_02 experiment. Overall treatment does not affect CosMx quality.

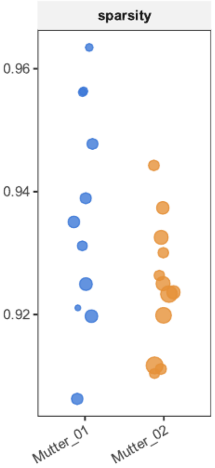
Bias and Between Replicate Concordance



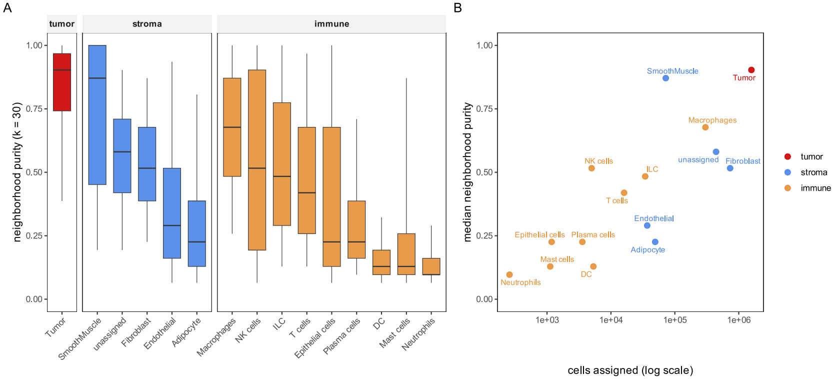
PCA of z-scored technical metrics show no pattern among replicates. Pseudobulk of all sample cells show within-cohort concordance.

Analysis Updates: Cell Identity

- Original: Default CosMx
  - InSituType maximum likelihood vs CosMX ImmGen atlas
- New: Luecken & Theis, *Molecular Systems Biology* (2019)
  - Unsupervised clustering of unlabeled cells (scVI + Leiden kNN), *then* label the clusters
  - True tumor / stroma separation

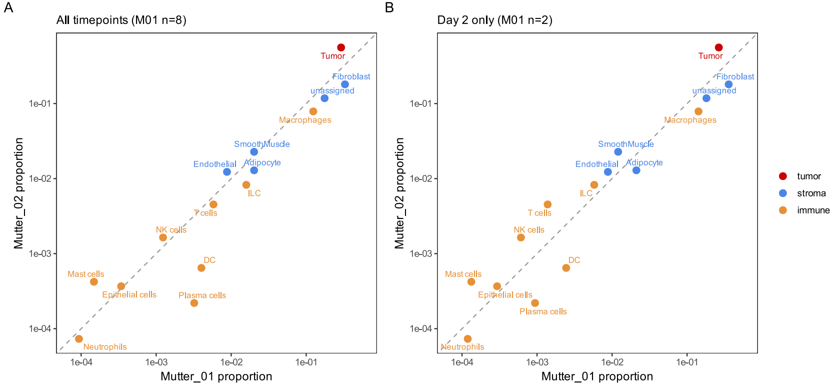


Analysis Updates: Cell Identity



Transcriptional neighborhood purity of the locked cell-type labels (fraction of k=30 scVI-latent neighbors sharing a cell's subtype). (A) Per-subtype purity by compartment; (B) median purity vs. cells assigned (log scale). Purity tracks positively with subtype abundance (Spearman rho ~0.82).

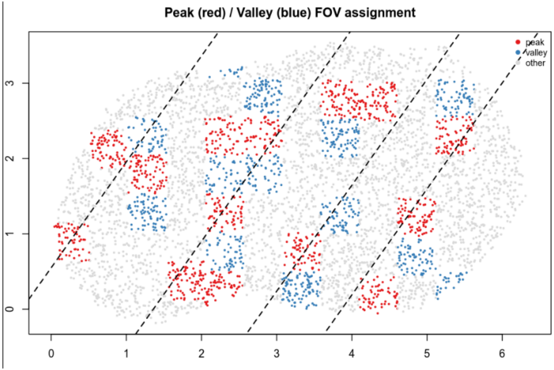
Analysis Updates: Cell Identity



Cell identity between experiments. (A) All time points. (B) Only at 2d.

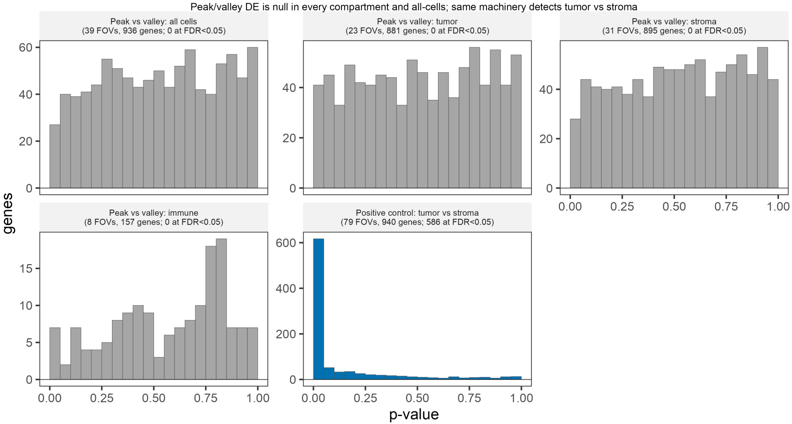
Analysis Updates: M_01 Peak and Valley

- ROI ID is still fine
- Original DGE: Pseudobulk per cell as replicates - inflated by very high replicate count Squair *et al.*, *Nature Communications* (2021)
- New: Merritt *et al.*, *Nature Biotechnology* (2020)
    - Per FOV
    - Upfront check of cell abundence
    - Narrow peak and valley bands



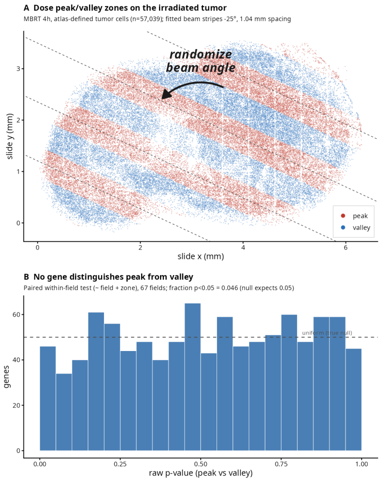
Results

No M_01 4hr Peak Valley Signature

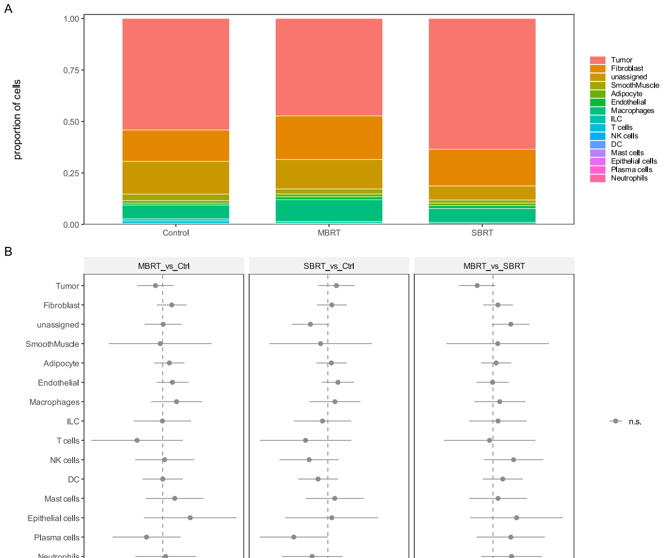


Peak-vs-valley FOV-pseudobulk DE at M_01 4h has no signature: p-value histograms are flat in all cells and every compartment.

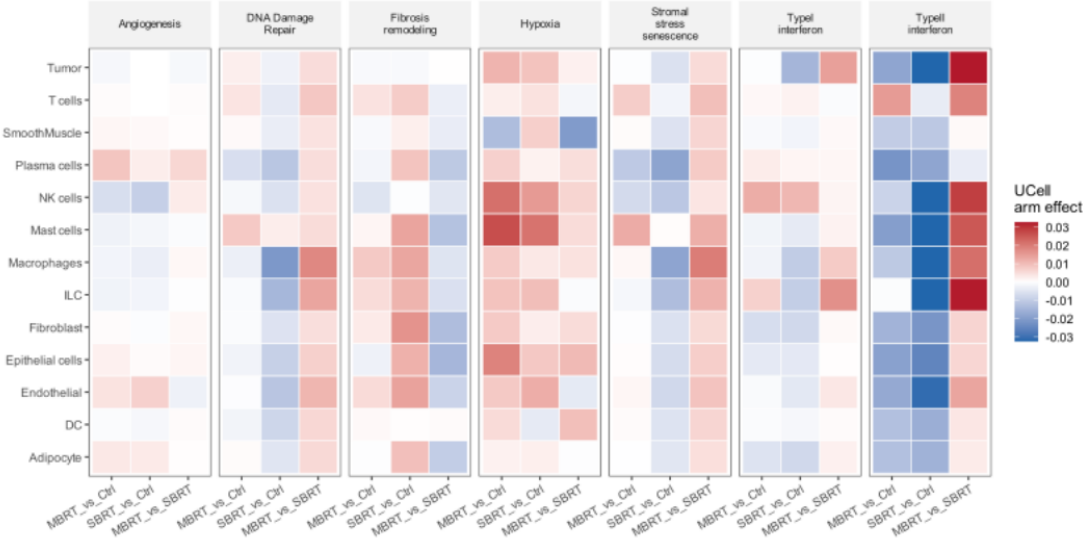
Alternative confirmation of peak valley signature loss



Arm-level Day 2 Results: Composition



Arm-level Day 2 Results: Pathway



Style reference

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FONTS

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Bold / alert	**10pt, bold**
Caption / cite	***10pt, bold, italic***

GEOMETRY

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